2.2 Complex Analysis: Study Notes

Notation: Ω is an open subset of $\mathbb{C}$; a *domain* (or *region*) is a non-empty open connected set. $D(a,r)$ is the open disc, $\mathbb{D} = D(0,1)$, $\mathbb{H} = \{\operatorname{Im} z > 0\}$.

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1. Holomorphic functions and the Cauchy–Riemann equations

Definition. $f : \Omega \to \mathbb{C}$ is *complex differentiable* at z_0 if

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0+h) - f(z_0)}{h}$$

exists, where $h \to 0$ through complex values. f is *holomorphic* on Ω if it is complex differentiable at every point of Ω. An *entire* function is holomorphic on $\mathbb{C}$.

Cauchy–Riemann. Write $f = u + iv$ with u, v real, $z = x + iy$. If f is complex differentiable at z_0, taking h real and then h purely imaginary gives

$$f'(z_0) = u_x + iv_x = \frac{1}{i}(u_y + iv_y),$$

so

$$u_x = v_y, \qquad u_y = -v_x.$$

Conversely, if u, v have *continuous* partial derivatives on Ω satisfying these equations, then f is holomorphic. (Continuity of partials gives real differentiability; CR says the real derivative, a 2×2 matrix, has the form $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$, i.e. it is multiplication by $a + ib$.)

Geometric meaning. At a point where $f'(z_0) \neq 0$, f acts infinitesimally as multiplication by $f'(z_0) = re^{i\theta}$: rotation by θ and scaling by r. So f preserves angles between curves (it is *conformal*).

Consequences. If f is holomorphic, u and v are harmonic ($\Delta u = 0$), granted that they are C^2 (which we will prove). With Wirtinger derivatives $\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$, $\partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y)$, CR is simply $\partial f / \partial \bar{z} = 0$.

Examples. Polynomials, $e^z = e^x(\cos y + i \sin y)$, $\sin z$, $\cos z$ are entire. $\bar{z}$ and $|z|^2$ are not holomorphic on any open set.

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2. Power series

Cauchy–Hadamard. $\sum a_n (z-a)^n$ has radius of convergence R given by $1/R = \limsup |a_n|^{1/n}$. It converges absolutely and locally uniformly in $D(a,R)$ and diverges for $|z-a| > R$.

Theorem. A power series is holomorphic inside its disc of convergence, and its derivative is the term-by-term derivative, which has the same radius of convergence. Hence power series are infinitely differentiable, and $a_n = f^{(n)}(a)/n!$.

The exponential and logarithm. $e^z = \sum z^n/n!$ is entire, $e^{z+w} = e^z e^w$, and e^z is $2\pi i$-periodic. On a simply connected domain not containing 0 there is a holomorphic *branch of the logarithm*: a holomorphic log with $e^{\log z} = z$. The *principal branch* on $\mathbb{C} \setminus (-\infty, 0]$ is $\operatorname{Log} z = \ln|z| + i \operatorname{Arg} z$ with $\operatorname{Arg} z \in (-\pi, \pi)$. Then $z^\alpha := e^{\alpha \log z}$.

Pitfall. $\log(zw) = \log z + \log w$ holds only modulo $2\pi i$. There is no continuous logarithm on $\mathbb{C} \setminus \{0\}$: going once around the origin changes arg by 2π.

3. Contour integrals and Cauchy's theorem

Definition. For a piecewise C^1 curve $\gamma : [a,b] \to \Omega$ and continuous f,

$$\int_\gamma f(z)\,dz = \int_a^b f(\gamma(t))\gamma'(t)\,dt.$$

ML-inequality: $\left|\int_\gamma f\right| \le \text{length}(\gamma) \cdot \sup_\gamma |f|$.

Primitives. If $F' = f$ on Ω then $\int_\gamma f = F(\gamma(b)) - F(\gamma(a))$. In particular the integral of f around any closed curve is 0.

Key example. For the circle C_r of radius r around a, traversed once anticlockwise,

$$\int_{C_r} (z-a)^n\,dz = \begin{cases} 2\pi i & n = -1, \\ 0 & n \ne -1. \end{cases}$$

(For $n \ne -1$, $(z-a)^{n+1}/(n+1)$ is a primitive.) Hence $1/(z-a)$ has no primitive on $\mathbb{C} \setminus \{a\}$.

Goursat's theorem. If f is holomorphic on Ω and $T \subseteq \Omega$ is a closed triangle (with interior), then $\int_{\partial T} f = 0$.

Proof sketch. Subdivide T into 4 congruent triangles by joining the midpoints of the sides. The integral over ∂T is the sum of the 4 integrals (interior edges cancel), so one of them, T_1, satisfies $|\int_{\partial T_1} f| \ge \frac{1}{4}|\int_{\partial T} f|$. Repeat to get nested triangles T_n with $|\int_{\partial T_n} f| \ge 4^{-n}|\int_{\partial T} f|$, diameter $d_n = 2^{-n}d$ and perimeter $p_n = 2^{-n}p$. They shrink to a point z_0 (compactness). Near z_0, $f(z) = f(z_0) + f'(z_0)(z-z_0) + \psi(z)(z-z_0)$ with $\psi(z) \to 0$. The linear part has a primitive so integrates to 0. Hence

$$\Big|\int_{\partial T_n} f\Big| \le p_n d_n \sup_{T_n} |\psi| = 4^{-n} p d \sup_{T_n} |\psi|.$$

Thus $|\int_{\partial T} f| \le pd \sup_{T_n} |\psi| \to 0$. □

Cauchy's theorem in a disc. If f is holomorphic in a disc D, then f has a primitive in D (define $F(z) = \int_{[c,z]} f$ along the segment from the centre c; Goursat shows $F(z+h) - F(z) = \int_{[z,z+h]} f$, so $F' = f$). Hence $\int_\gamma f = 0$ for every closed curve γ in D.

General versions. The same conclusion holds for any closed curve in a *simply connected* domain, and more generally (*homology version*) for any closed curve γ in Ω with winding number $n(\gamma, a) = 0$ for every $a \notin \Omega$. A useful practical form: if f is holomorphic on an open set containing a closed region bounded by finitely many piecewise smooth curves, the integral over the (positively oriented) boundary is zero. "Keyhole" and "indented" contour arguments use this.

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4. Cauchy's integral formula and its consequences

Cauchy's integral formula. Let f be holomorphic on an open set containing the closed disc $\overline{D(a,r)}$, and C its boundary circle (anticlockwise). Then for $|z-a| < r$,

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta.$$

Proof sketch. Apply Cauchy's theorem to the region between C and a small circle C_ε around z (a keyhole argument), so the integral over C equals the integral over C_ε. Write $\frac{f(\zeta)}{\zeta - z} = \frac{f(\zeta)-f(z)}{\zeta - z} + \frac{f(z)}{\zeta - z}$. The first term is bounded near z, so its integral over C_ε tends to 0; the second gives $2\pi i f(z)$. □

Holomorphic ⇒ analytic. Expand $\frac{1}{\zeta - z} = \sum_{n \geq 0} \frac{(z-a)^n}{(\zeta - a)^{n+1}}$ (uniformly for $\zeta \in C$, fixed $|z - a| < r$) and integrate term by term:

$$f(z) = \sum_{n \geq 0} a_n (z-a)^n, \qquad a_n = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - a)^{n+1}}\, d\zeta = \frac{f^{(n)}(a)}{n!}.$$

So a holomorphic function is infinitely differentiable, and its Taylor series at a converges on the largest disc around a contained in Ω.

Cauchy estimates. If $|f| \leq M$ on C, then $|f^{(n)}(a)| \leq n!\, M / r^n$.

Liouville's theorem. A bounded entire function is constant. (Cauchy estimate with $n = 1$ and $r \to \infty$ gives $f'(a) = 0$.)

Fundamental theorem of algebra. A non-constant polynomial p has a root in $\mathbb{C}$. (If not, $1/p$ is entire and tends to 0 at ∞, so it is bounded, hence constant.)

Morera's theorem. If f is continuous on a disc and $\int_{\partial T} f = 0$ for every triangle T in it, then f is holomorphic. (The proof of Cauchy in a disc gives a primitive F, which is holomorphic, hence so is $F' = f$.)

Uniform limits. If f_n are holomorphic on Ω and $f_n \to f$ uniformly on compact subsets, then f is holomorphic and $f_n' \to f'$ uniformly on compact subsets. (Morera for the first part, Cauchy's formula for the derivative for the second.) This is false in real analysis: $\sqrt{x^2 + 1/n} \to |x|$ uniformly.

Identity theorem. Let f be holomorphic on a domain Ω. If the zero set of f has a limit point in Ω, then $f \equiv 0$.

Proof sketch. Let U be the set of points where all derivatives of f vanish. U is closed (intersection of closed sets). U is open (power series). If $z_0 \in \Omega$ is a limit point of zeros and $f \not\equiv 0$ near z_0, write $f(z) = (z - z_0)^m g(z)$ with $g(z_0) \neq 0$; then $f \neq 0$ in a punctured neighbourhood, contradiction. So $z_0 \in U$, and $U \neq \emptyset$; connectedness gives $U = \Omega$. □

Consequence: two holomorphic functions on a domain that agree on a set with a limit point in the domain agree everywhere. Example: $\sin^2 z + \cos^2 z = 1$ on $\mathbb{C}$ because it holds on $\mathbb{R}$.

Pitfall. The limit point must lie *in* Ω. $\sin(1/z)$ on $\mathbb{C} \setminus \{0\}$ vanishes at $1/(k\pi)$, which accumulate only at $0 \notin \Omega$.

Mean value property. $f(a) = \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{it})\, dt$.

Maximum modulus principle. If f is holomorphic on a domain Ω and $|f|$ has a local maximum in Ω, then f is constant. Hence on a bounded domain, if f is continuous on $\overline{\Omega}$, $\max_{\overline{\Omega}} |f|$ is attained on $\partial\Omega$.

Proof sketch. If $|f(a)| \geq |f|$ on $\overline{D(a, r)}$, the mean value property gives $|f(a)| \leq \frac{1}{2\pi} \int |f(a + re^{it})| dt \leq |f(a)|$, forcing $|f| = |f(a)|$ on each such circle, so $|f|$ is constant near a, so f is constant near a (CR, or open mapping), and then on Ω by the identity theorem. □

Open mapping theorem. A non-constant holomorphic function on a domain maps open sets to open sets. (Proved with the argument principle, Section 7.)

Schwarz lemma. If $f : \mathbb{D} \to \mathbb{D}$ is holomorphic and $f(0) = 0$, then $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$. If equality $|f(z)| = |z|$ holds for some $z \neq 0$, or $|f'(0)| = 1$, then $f(z) = e^{i\theta} z$.

Proof. $g(z) = f(z)/z$ (with $g(0) = f'(0)$) is holomorphic on $\mathbb{D}$. On $|z| = r < 1$, $|g| \le 1/r$; by maximum modulus $|g| \le 1/r$ on $D(0,r)$; let $r \to 1$. If $|g| = 1$ at an interior point, g is a constant of modulus 1. □

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5. Laurent series and isolated singularities

Laurent's theorem. If f is holomorphic on the annulus $A = \{r < |z-a| < R\}$ ($0 \le r < R \le \infty$), then

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-a)^n, \qquad c_n = \frac{1}{2\pi i} \int_{|\zeta - a| = \rho} \frac{f(\zeta)}{(\zeta - a)^{n+1}}\, d\zeta \quad (r < \rho < R),$$

converging absolutely and locally uniformly on A. The expansion is unique, so any method of finding it (geometric series, known Taylor series) gives "the" Laurent series. **The series depends on the annulus:** $1/(z(z-1))$ has different expansions on $0 < |z| < 1$ and $|z| > 1$.

Isolated singularities. Suppose f is holomorphic on $D(a,R) \setminus \{a\}$ with Laurent series $\sum c_n (z-a)^n$.

Type	Laurent series	Behaviour near a
Removable	$c_n = 0$ for all $n < 0$	f bounded near a (Riemann's theorem: bounded ⇒ removable)
Pole of order m	$c_{-m} \neq 0$, $c_n = 0$ for $n < -m$	$\lvert f(z)\rvert \to \infty$; $f = g/(z-a)^m$ with $g(a) \neq 0$
Essential	$c_n \neq 0$ for infinitely many $n < 0$	**Casorati–Weierstrass:** $f(D(a,\varepsilon) \setminus \{a\})$ is dense in $\mathbb{C}$ for every ε

(Great Picard theorem, statement only: near an essential singularity f takes every complex value, with at most one exception, infinitely often.)

Proof of Casorati–Weierstrass. If $|f(z) - w| \ge \delta$ on a punctured disc, then $g = 1/(f-w)$ is bounded there, so has a removable singularity. Then $f = w + 1/g$ has a removable singularity or a pole at a. □

A function is *meromorphic* on Ω if it is holomorphic except at isolated poles. Meromorphic functions on the Riemann sphere $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ are exactly rational functions. (Behaviour at ∞ is defined via $g(w) = f(1/w)$ at $w = 0$.)

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6. Residues

Definition. The *residue* of f at an isolated singularity a is $\operatorname{Res}(f,a) = c_{-1}$, the coefficient of $(z-a)^{-1}$.

Residue theorem. Let f be holomorphic on an open set containing a closed disc (or any closed region with a positively oriented piecewise smooth boundary γ), except at finitely many points $a_1, \dots, a_k$ inside. Then

$$\int_\gamma f(z)\, dz = 2\pi i \sum_j \operatorname{Res}(f, a_j).$$

(Subtract principal parts; only the $(z-a_j)^{-1}$ terms survive integration.)

Computing residues.

- Simple pole: $\operatorname{Res}(f,a)=\lim_{z\to a}(z-a)f(z)$. If $f=g/h$ with $g(a)\neq 0$, $h(a)=0$, $h'(a)\neq 0$: Res $=g(a)/h'(a)$.
- Pole of order m: $\operatorname{Res}(f,a)=\dfrac{1}{(m-1)!}\lim_{z\to a}\dfrac{d^{m-1}}{dz^{m-1}}\big[(z-a)^m f(z)\big]$.
- Essential singularity: expand the Laurent series directly (for example $e^{1/z}$ has residue 1 at 0).

Standard real integrals.

1. *Rational functions:* $\int_{-\infty}^{\infty} P(x)/Q(x)\,dx$ with $\deg Q\geq \deg P+2$ and no real zeros of Q: close in the upper half-plane with a semicircle of radius R; the arc contributes $O(1/R)$. Example: $\int_{-\infty}^{\infty}\frac{dx}{1+x^2}=2\pi i\cdot\frac{1}{2i}=\pi$.
2. *Trigonometric integrals:* $\int_0^{2\pi} R(\cos\theta,\sin\theta)\,d\theta$: put $z=e^{i\theta}$, $\cos\theta=\frac{z+z^{-1}}{2}$, $\sin\theta=\frac{z-z^{-1}}{2i}$, $d\theta=\frac{dz}{iz}$, and integrate over $|z|=1$.
3. *Fourier-type integrals:* $\int_{-\infty}^{\infty} f(x)e^{ix}\,dx$ with $f\to 0$: use **Jordan's lemma**: on the upper semicircle C_R,
$$\Big|\int_{C_R} g(z)e^{iz}\,dz\Big|\leq \pi\max_{C_R}|g|,$$
because $\int_0^{\pi} e^{-R\sin\theta}\,d\theta<\pi/R$.
4. *Indented contours:* for a simple pole on the real axis, go around it on a small semicircle of radius ε; as $\varepsilon\to 0$ the small semicircle contributes $-\pi i\operatorname{Res}$ (clockwise, half the full-circle value). Gives $\int_0^{\infty}\frac{\sin x}{x}\,dx=\frac{\pi}{2}$.
5. *Keyhole contours:* for $\int_0^{\infty} x^{a-1}R(x)\,dx$ or integrals with $\log x$: cut along $[0,\infty)$, use the branch with arg $\in(0,2\pi)$; the two sides of the cut differ by the factor $e^{2\pi i(a-1)}$.

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7. The argument principle and Rouché's theorem

Argument principle. Let f be meromorphic on an open set containing a closed disc $\overline{D}$, with no zeros or poles on ∂D. Then
$$\frac{1}{2\pi i}\int_{\partial D}\frac{f'(z)}{f(z)}\,dz=Z-P,$$
where Z and P are the numbers of zeros and poles inside D, counted with multiplicity. (Near a zero of order m, $f'/f=\frac{m}{z-a}+\text{holomorphic}$; near a pole of order m, $f'/f=\frac{-m}{z-a}+\text{holomorphic}$.) Geometrically, $Z-P$ is the winding number of the curve $f(\partial D)$ around 0.

Rouché's theorem. Let f,g be holomorphic on an open set containing $\overline{D}$, and suppose $|g(z)|<|f(z)|$ for all $z\in\partial D$. Then f and $f+g$ have the same number of zeros in D (with multiplicity).

Proof sketch. For $t\in[0,1]$, $f_t=f+tg$ has no zeros on ∂D (since $|f|>|g|\geq t|g|$). The number of zeros $\frac{1}{2\pi i}\int f_t'/f_t$ is integer-valued and continuous in t, hence constant. □

"Dog on a leash": if the leash $|g|$ is shorter than the distance $|f|$ of the person from the lamppost, the dog and the person walk around the lamppost the same number of times.

Consequences.

- *Fundamental theorem of algebra* (again): z^n dominates $a_{n-1}z^{n-1}+\cdots+a_0$ on a large circle, so p has n zeros.
- *Open mapping theorem*: if $f(a)=w_0$, then for w close to w_0, Rouché applied to $f-w_0$ and the constant w_0-w on a small circle around a shows $f-w$ has a zero near a.
- *Hurwitz's theorem*: if $f_n\to f$ locally uniformly on a domain, f_n have no zeros, then f has no zeros or $f\equiv 0$. If the f_n are injective, f is injective or constant. (Used in the Riemann mapping theorem.)
- *Local behaviour*: if $f(a)=w_0$ with $f-w_0$ having a zero of order m at a, then f is exactly m-to-1 near a (onto a punctured neighbourhood of w_0). In particular, injective holomorphic maps have non-vanishing derivative, and their inverses are holomorphic.

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8. Conformal maps and Möbius transformations

Conformal equivalence. A *conformal map* (biholomorphism) between open sets U, V is a holomorphic bijection $U\to V$; its inverse is automatically holomorphic.

Möbius transformations. $T(z)=\dfrac{az+b}{cz+d}$ with $ad-bc\neq 0$, viewed as a bijection of $\hat{\mathbb{C}}$ ($T(-d/c)=\infty$, $T(\infty)=a/c$).

- They form a group under composition, isomorphic to $\mathrm{PGL}_2(\mathbb{C})=\mathrm{GL}_2(\mathbb{C})/\mathbb{C}^*$ (compose $\leftrightarrow$ multiply matrices).
- Every Möbius map is a composition of translations, rotation-dilations $z\mapsto az$, and inversion $z\mapsto 1/z$.
- They map "circles" (circles and straight lines; a straight line is a circle on $\hat{\mathbb{C}}$ through ∞) to circles.
- There is exactly one Möbius map sending three given distinct points z_1, z_2, z_3 to $0, 1, \infty$:

$$T(z)=\frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)} \quad \text{(the cross-ratio)}.$$

  So a Möbius map is determined by the images of three points.

Key examples.

- *Cayley transform:* $z\mapsto \dfrac{z-i}{z+i}$ maps $\mathbb{H}$ conformally onto $\mathbb{D}$, with inverse $w\mapsto i\dfrac{1+w}{1-w}$.
- $\mathrm{Aut}(\mathbb{D})=\left\{z\mapsto e^{i\theta}\dfrac{z-a}{1-\bar{a}z} : \theta\in\mathbb{R}, |a|<1\right\}$ (proof via Schwarz lemma; see Problem 22).
- $\mathrm{Aut}(\mathbb{H})=\left\{z\mapsto \dfrac{az+b}{cz+d} : a,b,c,d\in\mathbb{R}, ad-bc=1\right\}\cong \mathrm{PSL}_2(\mathbb{R})$.
- $\mathrm{Aut}(\mathbb{C})=\{az+b : a\neq 0\}$ (Problem 20).
- Other building blocks: $z\mapsto z^\alpha$ opens or closes sectors; e^z maps the strip $\{0<\mathrm{Im}\, z<\pi\}$ onto $\mathbb{H}$; log maps $\mathbb{H}$ back onto the strip.

Riemann mapping theorem (statement). If $\Omega\subsetneq\mathbb{C}$ is a non-empty simply connected open set and $z_0\in\Omega$, there is a unique conformal map $F:\Omega\to\mathbb{D}$ with $F(z_0)=0$ and $F'(z_0)>0$.

Outline of proof (SS Ch. 8). (1) Using a square root, conformally embed Ω into $\mathbb{D}$ with $z_0\mapsto 0$. (2) Consider the family $\mathcal{F}$ of injective holomorphic maps $\Omega\to\mathbb{D}$ fixing $z_0\mapsto 0$; it is non-empty and uniformly bounded. (3) **Montel's theorem** (a uniformly bounded family of holomorphic functions is normal; this is Arzelà–Ascoli plus Cauchy estimates) gives a limit f maximising $|f'(z_0)|$; Hurwitz shows f is injective. (4) If f missed a point of $\mathbb{D}$, composing with disc automorphisms and a square root would produce an element of $\mathcal{F}$ with larger derivative at z_0, contradicting maximality. Uniqueness follows from the Schwarz lemma.

$\mathbb{C}$ itself is excluded: by Liouville, there is no non-constant holomorphic map $\mathbb{C}\to\mathbb{D}$.

9. Common pitfalls

1. Complex differentiability at a single point is weak (e.g. $|z|^2$ at 0); holomorphy requires an open set.
2. Real-valued holomorphic functions on a domain are constant (CR gives $u_x = u_y = 0$; or open mapping).
3. Check that the contribution from large arcs really tends to zero; for e^{iz} use Jordan's lemma, for e^{-iz} close in the *lower* half-plane.
4. Orientation: the residue theorem uses positively oriented (anticlockwise) boundaries. Indentations traversed clockwise contribute with a minus sign.
5. Branch choices for log and z^α must be consistent along the whole contour; state the branch explicitly.
6. When counting with Rouché, the inequality must be strict on the *whole* boundary circle.
7. Residue at a pole of order ≥ 2 is not $g(a)/h'(a)$; use the derivative formula or a Laurent expansion.

10. Connections forward

- **2.4 Analysis in $\mathbb{R}^n$:** Arzelà–Ascoli is the engine behind Montel's theorem; Green's theorem and differential forms give another proof of Cauchy's theorem ($f\,dz$ is closed iff f is holomorphic).
- **3.1 Measure theory and 3.2 Functional analysis:** Fourier analysis and the Paley–Wiener theorems; Hardy spaces H^p; the spectrum of a bounded operator is non-empty because the resolvent is a bounded holomorphic function (Liouville); holomorphic functional calculus.
- **3.4 Algebraic topology:** winding numbers and the fundamental group of the circle; the argument principle is the degree of a map $S^1 \to S^1$; branches of log and covering spaces.
- **3.5 Smooth manifolds:** Riemann surfaces are complex 1-manifolds; the Riemann sphere $\widehat{\mathbb{C}} \cong \mathbb{CP}^1$.
- **Electives:** analytic number theory (the zeta function and the prime number theorem: SS Ch. 6–7), Riemann surfaces, several complex variables.